

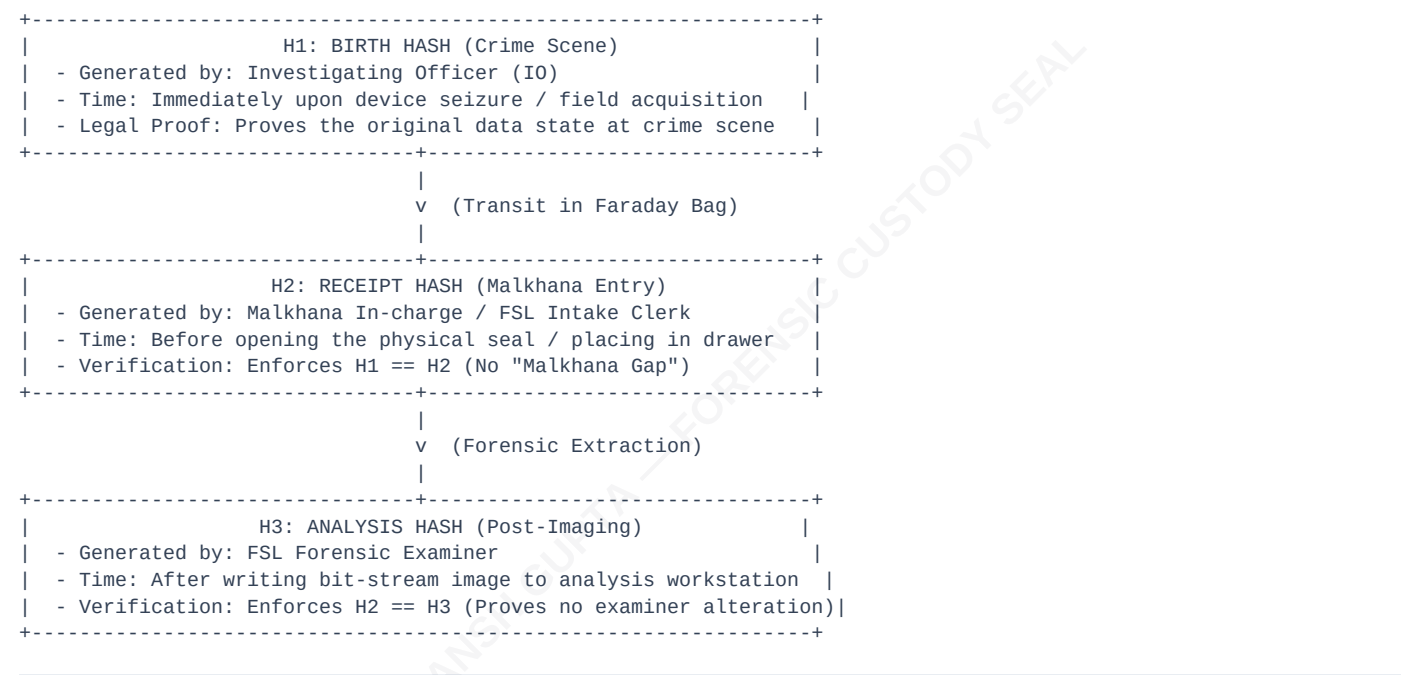
TRIPLE-HASH VERIFICATION PROTOCOL (H1 → H2 → H3)

Digital evidence is highly volatile. The primary challenge in court is proving that a seized electronic device has not been modified between the crime scene and the forensic laboratory.

To satisfy the stringent standards of **Section 63 of the Bharatiya Sakshya Adhiniyam (BSA), 2023**, Malkhana Vault implements the **Triple-Hash Verification Protocol**.

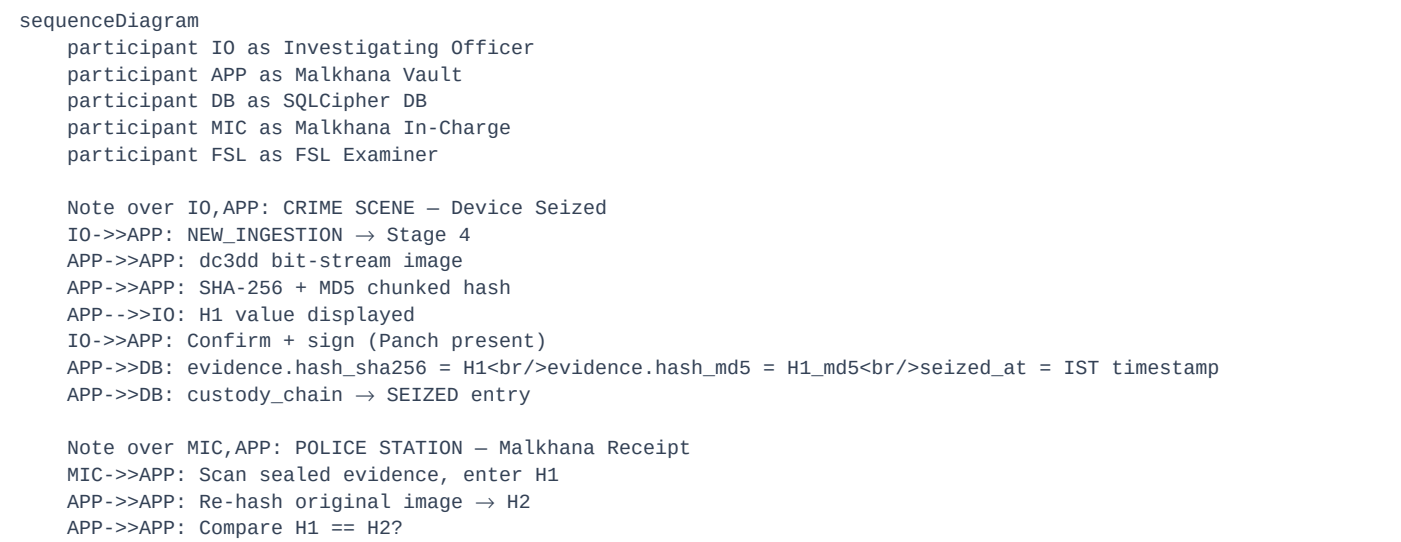
1. The Three Cryptographic Checkpoints

Malkhana Vault mandates dual-hashing (generating both **SHA-256** and **MD5** simultaneously) at three distinct milestones in the custody lifecycle:



2. Sequence Diagram

This sequence diagram illustrates how the three hashes are verified and tracked across roles:



```
alt H1 == H2 - Integrity confirmed
  APP-->>MIC: ■ MALKHANA ACCEPTED
  APP-->>DB: custody_chain → INTAKE<br/>hash_at_transfer = H2<br/>hash_verified = 1
  APP-->>DB: archive_matrix slot assigned
else H1 ≠ H2 - Malkhana Gap detected
  APP-->>MIC: ■ HASH MISMATCH - MALKHANA GAP<br/>Evidence may have been accessed in transit
  APP-->>DB: custody_chain → INTAKE<br/>hash_verified = 0<br/>notes = MISMATCH FLAGGED
  Note over APP,DB: Certificate engine will<br/>include gap warning
end

Note over FSL,APP: FSL LAB - Analysis
FSL-->>APP: Receive sealed device, verify H2
APP-->>APP: Re-hash on FSL machine → H3
APP-->>APP: Compare H2 == H3?

alt H2 == H3 - No FSL tampering
  APP-->>FSL: ■ ANALYSIS AUTHORISED
  APP-->>DB: custody_chain → EXAMINED<br/>hash_at_transfer = H3<br/>hash_verified = 1
else H2 ≠ H3 - FSL alteration detected
  APP-->>FSL: ■ HASH MISMATCH - FSL ALTERATION<br/>Forensic examiner may have modified data
  APP-->>DB: custody_chain → EXAMINED<br/>hash_verified = 0<br/>notes = FSL MISMATCH
end

FSL-->>APP: Generate Section 63 certificate
APP-->>DB: Read all custody_chain entries
APP-->>APP: Build certificate with hash audit trail
APP-->>FSL: PDF ready for court submission
```

3. Legal Implications of Mismatches

- **H1 ≠ H2 ("Malkhana Gap"):** Points to tampering or access during transport from the crime scene to the station. Under BNSS Section 153 and BSA Section 63, this mismatch is grounds for the defense to challenge the admissibility of the electronic record.
- **H2 ≠ H3 ("Examiner Alteration"):** Indicates that forensic extraction altered the source evidence (e.g. failure to use a hardware write-blocker).
- **The Vault Safeguard:** Malkhana Vault is an **append-only database**. H1, H2, and H3 values cannot be overwritten or edited once written. All mismatches are highlighted automatically in the generated BSA Section 63 Certificate, ensuring transparent disclosure before the court.